Aritra Basak

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Software Engineer | Backend Developer

Software Engineer with 3 years of experience designing and scaling high-concurrency microservices using Java, Spring Boot, and AWS. Proven track record of delivering robust applications, modernizing legacy systems, integrating AI features and optimizing performance for large-scale enterprise and government clients.

Education

Narula Institute of Technology, B-Tech in Electronics and Communication Engineering | 08/2019 – 06/2023

- DGPA: 9.07/10

Julien Day School, Class 12 (ISC) | 04/2018 – 06/2019

- Percentage: 81/100

Julien Day School, Class 10 (ICSE) | 04/2016 – 06/2017

- Percentage: 84/100

Experience

Consultant, EY – Kolkata, India | 07/2023 – Present

- Led the revamp of a legacy application into a microservices-based architecture using Spring Boot, gRPC, and JWT Auth for a West Bengal Government initiative.
- Lead Engineer for the Ayushman Bharat Digital Mission Connector v2.0, directing full-stack development (React.js, Spring Boot) and collaborating directly with government clients to ensure technical alignment with project requirements.
- Engineered a multifunctional Insurance Claim Details Fetching Chatbot using Python, integrating LangGraph, Ollama, PGVector, and all-MiniLM-L6-v2 embeddings; orchestrated multiple specialized agents for arithmetical computation, analytical reasoning, query filtering, and vector-based semantic retrieval.
- Developed an AI-driven Grievance Portal using GenAI models for multilingual text extraction from grievance forms for the Chhattisgarh Government, reducing manual data processing time by 60% and accelerating response times by 2× for the Chhattisgarh Government.
- Developed an LLM-integrated Python automation system for the West Bengal Birth and Death Portal, enabling intelligent document classification and data extraction from uploaded birth and death certificates, increasing validation accuracy by 35% and reducing manual verification time by 80%.
- Worked directly with government clients from West Bengal, Telangana, and Rajasthan on software implementation projects, contributing to public sector digital transformation through hands-on R&D and solution delivery.
- Engineered a Python script for medical insurance fraud detection, significantly augmenting the system's ability to identify fraudulent claims by an impressive 70%.
- Developed a standalone Java service using multithreading and concurrency to optimize health record linking, increasing daily processing volume by over 100% for a West Bengal Government project.

Java Programmer Intern , Suven Consultants and Technologies Pvt. Ltd. – Kolkata, India | 07/2022 – 08/2022

- Strengthened Java programming skills through hands-on project work, developing Java projects based on core principles to reinforce learning.

Projects

Secure Peer-to-Peer UPI Wallet System. GitHub URL

- **T**he Secure Peer-to-Peer UPI Wallet System is a banking-grade REST API that simulates a digital wallet with a strict double-entry ledger to guarantee ACID compliance. It leverages pessimistic database locking to prevent "double-spend" race conditions during concurrent transfers, and implements the Saga Orchestrator Pattern to handle distributed network failures with automated refunds.
- Tools Used: Java 17, Spring Boot, Spring Data JPA, MySQL, SLF4J (MDC), Maven, Git .

High-Concurrency User Service with Fault-Tolerance. GitHub URL

- **B**uilt a Spring Boot POC demonstrating real-world concurrent user handling using optimistic locking, Resilience4j (circuit breaker, rate limiting, bulkhead, retry), Caffeine caching, and multithreaded integration testing.
- Tools Used: Java, Spring Boot 3, Spring Data JPA, Hibernate, H2 Database, Resilience4j, Caffeine Cache, Spring Cache, Spring Boot Actuator, JUnit 5, Maven, Git .

Big Brain. Project URL

- Developed and deployed a functional application enabling users to upload, query, and perform vector searches on documents and notes while facilitating collaborative sharing through organizations.
- Tools Used: Next.Js, Typescript, Clerk, Convex, Groq-Cloud Inference, Shadecn UI.

Graph QL Integration. GitHub URL

- **O**ptimized API performance by migrating from REST to GraphQL, achieving a 45% latency reduction and allowing users to dynamically select fields through parameterized queries for more efficient data retrieval.
- Tools Used: Java, Spring Boot, Graph QL, Postman API.

Technologies

Languages: Java, Python, JavaScript, GraphQL

Frameworks and Libraries: Spring Boot, Spring MVC, Spring Data, Spring Security, React.js, Tailwind CSS, Bootstrap, LangGraph

Cloud and DevOps: AWS (EC2, S3, Lambda, Kinesis, IAM, Bedrock, CloudWatch), Apache APISIX, Docker, Git, Postman, Swagger, Grafana

Architecture and Patterns: REST API, gRPC, Prompt Engineering, RAG pipelines, Monolithic and Microservices Architecture

Databases: SQL, MongoDB

Honors and Awards

Client Extraordinaire 2024

Recognized as 'Client Extraordinaire' for delivering exceptional technical results and maintaining a good client satisfaction rate across major projects.

Emerging Extraordinaire 2023

Received "Emerging Extraordinaire" award for outstanding performance during the first six months at EY